

Shape of cubics



1 Yes

2

- a they become very large in the positive direction
- b they become very large in the negative direction

3 Odd, because the ends of the function point in opposite directions.

4

- a $x < 0$
- b $x > 0$
- c $(0, 0)$
- d The point of inflection

5

- a $x > -3$
- b $x < -3$
- c $(-3, 0)$

6 y increases more slowly on $y = \frac{1}{2}x^3$ than on $y = x^3$

7 $x = -1, \quad x \geq 3$

8

- a $(0, 0)$
- b $(0, 3)$
- c $(0, 4)$

Finding intercepts and equations of cubics

9

- a Negative
- b $(0, 3)$
- c $x < 0$
- d $x > 0$
- e $(0, 3)$

10

- a No
- b No
- c No
- d Yes

11

- a $x = -2$
- b $y = 4$

12 3

13

a negative

b $(0, 3)$ 

c A

14 $y = (x - 10)^3$

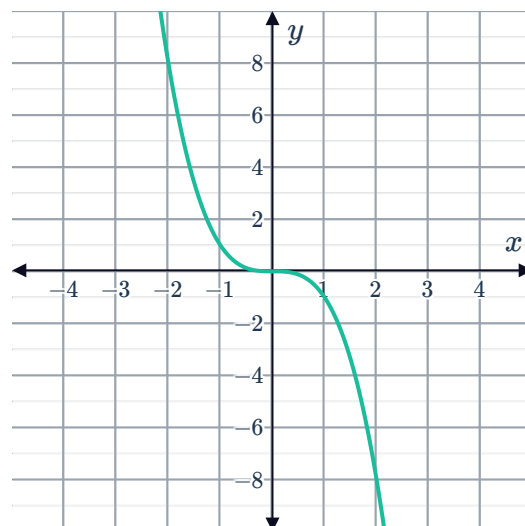
Graph of cubics

15

a

x	-2	-1	0	1	2
y	8	1	0	-1	-8

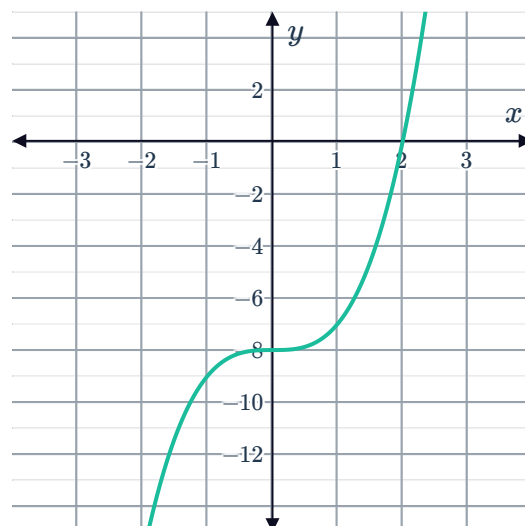
b



16

a $x = 2$ c $(0, -8)$ b $y = -8$

d



17



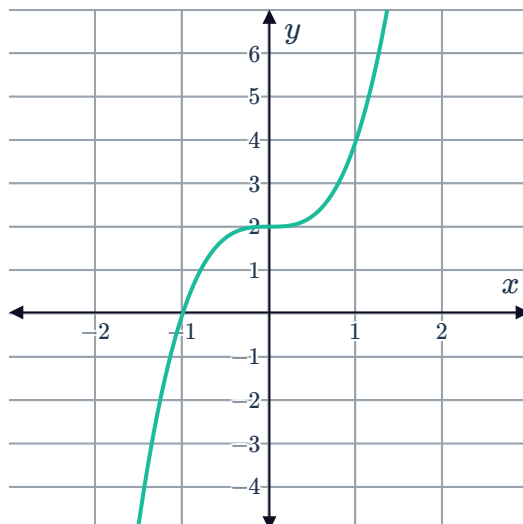
a

i Increasing

iii $(0, 2)$

ii More steep

iv



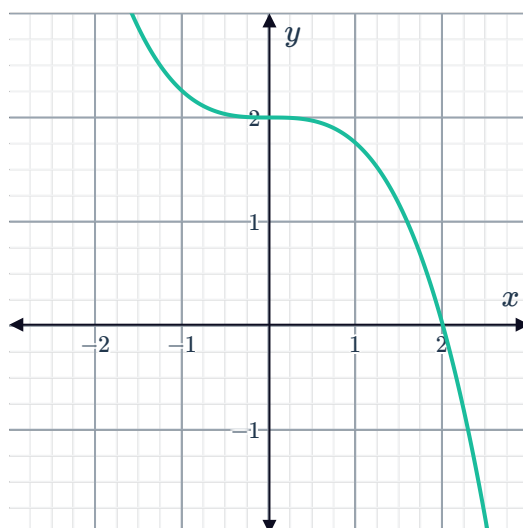
b

i Decreasing

iii $(0, 2)$

ii Less steep

iv



c

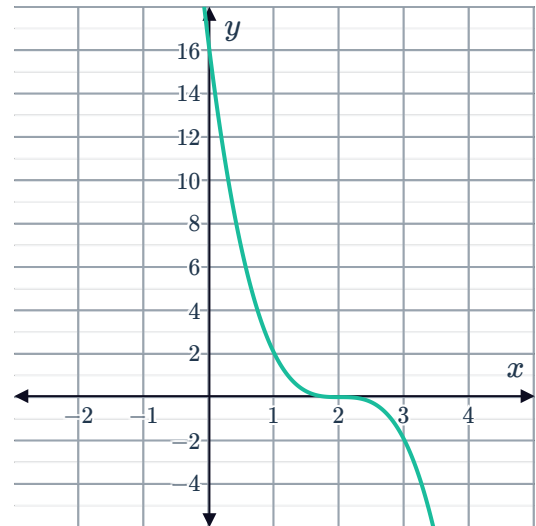
i Decreasing

iii $(2, 0)$



ii More steep

iv



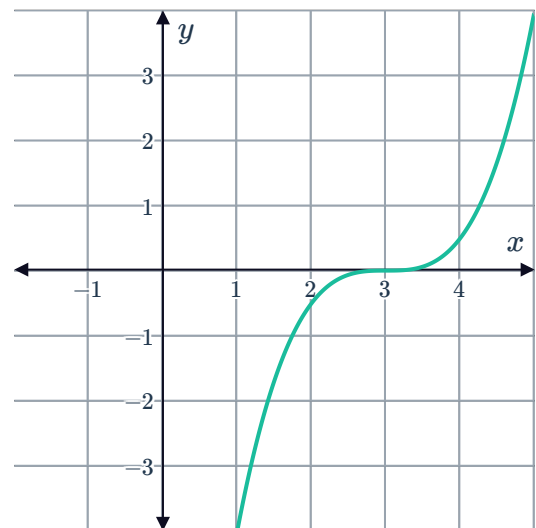
d

i Increasing

iii $(3, 0)$

ii Less steep

iv



e

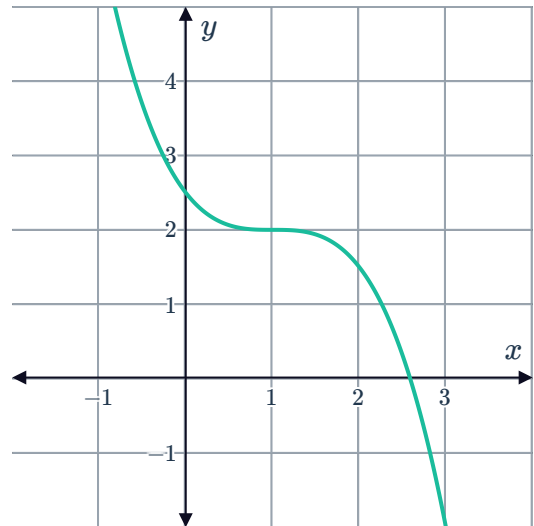
i Decreasing

iii (1,2)



ii Less steep

iv



18

a $x = 2$

c (1,3)

b $y = 6$

d

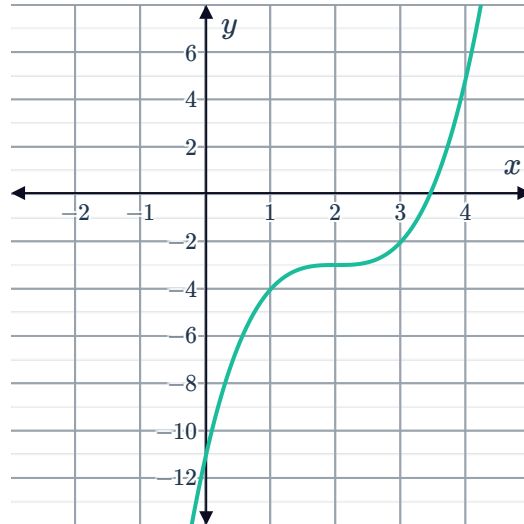


19

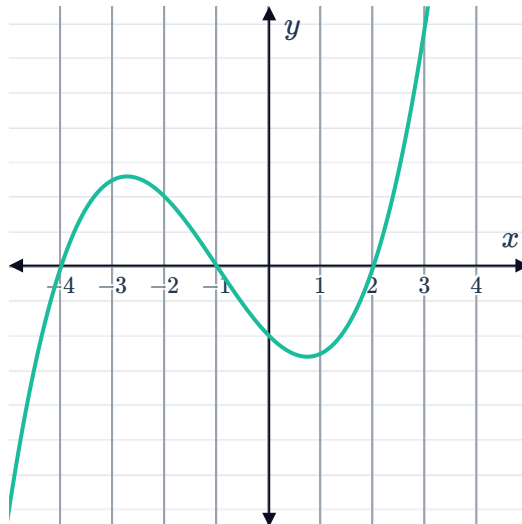


a Move the graph to the right by 2 units and down by 3 units.

b



20



21

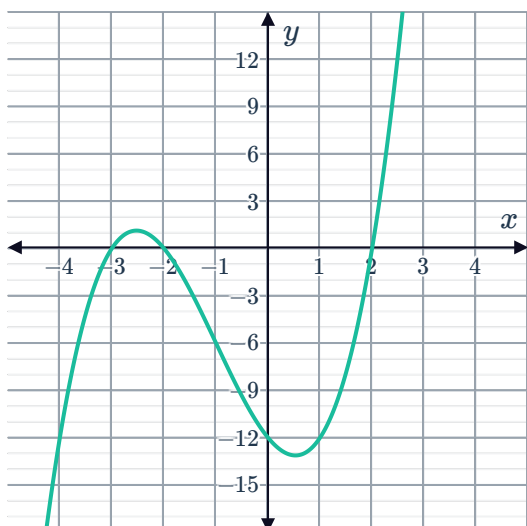


a

i $x = -3, -2, 2$

ii $y = -12$

iii

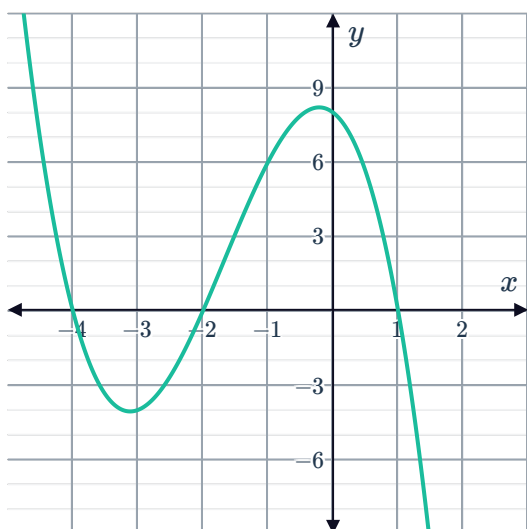


b

i $x = -4, -2, 1$

ii $y = 8$

iii



22

a $y = -x(x - 3)(x + 1)$

c $x = 0, 3, -1$



b $y = 0$

d

